12 · Cost Management Plan

The single most important financial fact in this whole handoff: **today the platform costs almost nothing because AWS credits cover it — in the Institute's own account, that becomes real money, and almost all of it is AI inference.** This document explains where the cost comes from, what drives it up or down, and the concrete levers and day-one steps to keep it under control.

Figures here are the **real billing numbers** from the audit (01 · Project Audit §3, AWS Cost Explorer, Jan–Aug 2026); this document turns them into a forward plan.

Companion documents: 01 · Project Audit (the raw cost report), 03 · Engine Comparison (per-engine cost), 09 · Product Roadmap (cheap-infra wins). Glossary in the README.

Part A · The headline

- **Gross AWS usage, Jan–Aug 2026: ≈ $371.** Net out-of-pocket: **effectively zero** (a few cents) — because the shared/university account's bill is covered by **AWS credits.**
- **The Institute's own account will not have those credits.** That ~$371 of usage becomes real money.
- **≈ 98% of the cost is AI (Amazon Bedrock LLM) inference** — reading documents and generating briefings. Everything else — servers, database, storage, delivery — totaled **~$7 over seven months (~$1/month)** and fits inside AWS's always-free allowances at this scale.

The one-sentence model: *cost scales with how much AI extraction and generation you run — not with uptime.* An idle platform costs about a dollar a month; a platform re-processing large corpora costs real money.

Part B · Where the money goes

From the real billing data (Jan–Aug 2026, gross):

Category	Gross	Share	What it is
Bedrock LLM inference (Claude Haiku/Sonnet, Llama, Titan, BDA)	**~$364**	**~98%**	Document extraction, labeling, eval loops, brief generation
Amazon Textract	$0.63	<1%	OCR (low — was SCP-blocked in the sandbox)
EC2/VPC (one-off)	~$4.37	~1%	A corpus/embedding build box

Category	Gross	Share	What it is
DynamoDB / S3 / Config / CloudWatch / CloudFront / SQS	<$3 total	~1%	The serverless infrastructure

And it is bursty. Of the ~$371, **$366 was a single month (July)** — the capstone crunch of re-running extraction over corpora and eval loops. Months with no development ran **from ~$0.14 (Mar–Jun combined) to ~$4/month.** This is *development* volume, not steady-state serving.

Part C · What it will cost the Institute (planning anchors)

Grounded in the measured usage:

Scenario	Cost	Notes
Idle (running, occasional demo, no bulk extraction)	**~$1–3/month**	Non-AI infra rate; fits free tiers; Bedrock ≈ $0 when not extracting
Cold storage (compute torn down, data kept)	**~$1/month**	S3 + a database export
Torn down entirely	**$0**	`sst remove` (note: production is `retain`, so its buckets survive and need manual cleanup)
Active development / bulk re-extraction	**~$50–350/month**	The measured heavy-dev range
Steady serving to end users (no bulk re-extraction)	**a fraction of the dev range**	Variable cost is dominated by extraction volume

The variable cost is **entirely LLM inference**, proportional to how many documents are extracted and how many briefings are generated.

Part D · The levers (how to control it)

Biggest lever — control extraction/generation volume: - **Extract on demand, not in bulk.** The costly pattern is re-running extraction across whole corpora. Steady-state serving of already-extracted data is cheap. - **Choose the cheaper engine where appropriate.** Per the engine comparison (03), on the same 8-document test the engines cost **~$2.13 (text-layer) vs. ~$3.44 (Textract-LAYOUT) vs. ~$7.92 (BDA)** — BDA is ~3.7× the cheapest. Text-layer is the residency-friendly, lowest-cost option for born-digital PDFs; Textract-LAYOUT is the recommended default for quality. Matching engine to need is a direct cost lever. - **Cache / avoid re-work.** Don't re-extract unchanged documents; the platform already avoids overwriting a RAP once progress is recorded.

Cheap infrastructure wins (small config changes, from 09 · Roadmap): - **Run Lambdas on arm64** (Graviton) — roughly **20% cheaper** compute. - **Add S3 lifecycle rules** (expire old uploads/exports) and **DynamoDB TTL** on staging rows. - **Reduce CloudWatch log retention** from the 30-day default where not needed.

These infra levers matter less in absolute dollars (~$1/month base) but are free wins and good hygiene as traffic grows.

Part E · Day-one actions (do these when you take the account)

1. **Set an AWS Budget alarm immediately** — e.g. alert at $25, $50, $100/month. This is the single most important safeguard; it turns a runaway extraction job into an email, not a surprise invoice.
2. **Confirm whether any credits apply** to the Institute's account (most won't have the sandbox's) — and plan for the real rate if not.
3. **Watch the first real extraction run** and note its cost, so you have a per-document anchor for your own usage.
4. **Turn on Cost Explorer** and check it monthly (there is **no in-app cost dashboard** — cost monitoring is done in the AWS console; see 08 · Content Stewardship §H).
5. **Right-size before scaling** — apply the arm64 / lifecycle / TTL wins (Part D) before any large corpus run.

Part F · What to watch as the platform grows

- **Bulk corpus ingestion is the cost event to plan for.** If the Institute decides to extract a large library of real RAPs (see 09 · Roadmap), model that as a **one-time project cost**, estimate it from a small pilot run first, and set a budget for it — don't run it open-ended.
- **Briefing/Q&A generation** also consumes inference; heavy use of the legal-cases assistant adds variable cost.
- **SES email** (the digest) is negligible at this scale but requires production access (see 09).
- **Steady-state serving stays cheap** — the serverless stack scales to near-zero when idle, which is the platform's cost advantage.

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*This plan is anchored in real billing data (net ≈ $0 in the credit-covered sandbox; ~$371 gross, ~98% Bedrock inference, with $366 of it a single development month). The takeaway for the Institute: budget for **AI inference proportional to how much you extract/generate**, protect the account with a Budget alarm on day one, and treat any bulk re-extraction as a planned, estimated project rather than a background activity.*